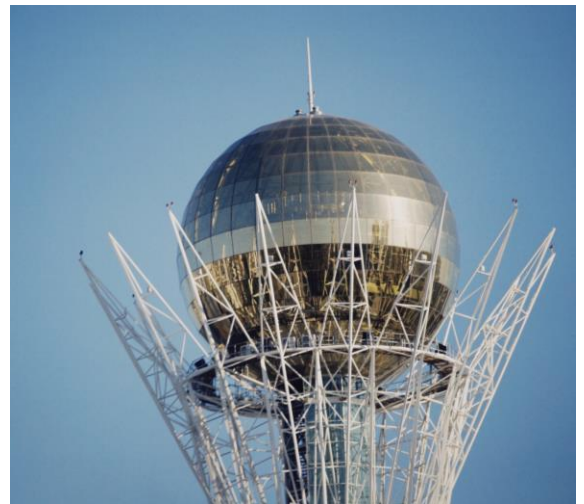
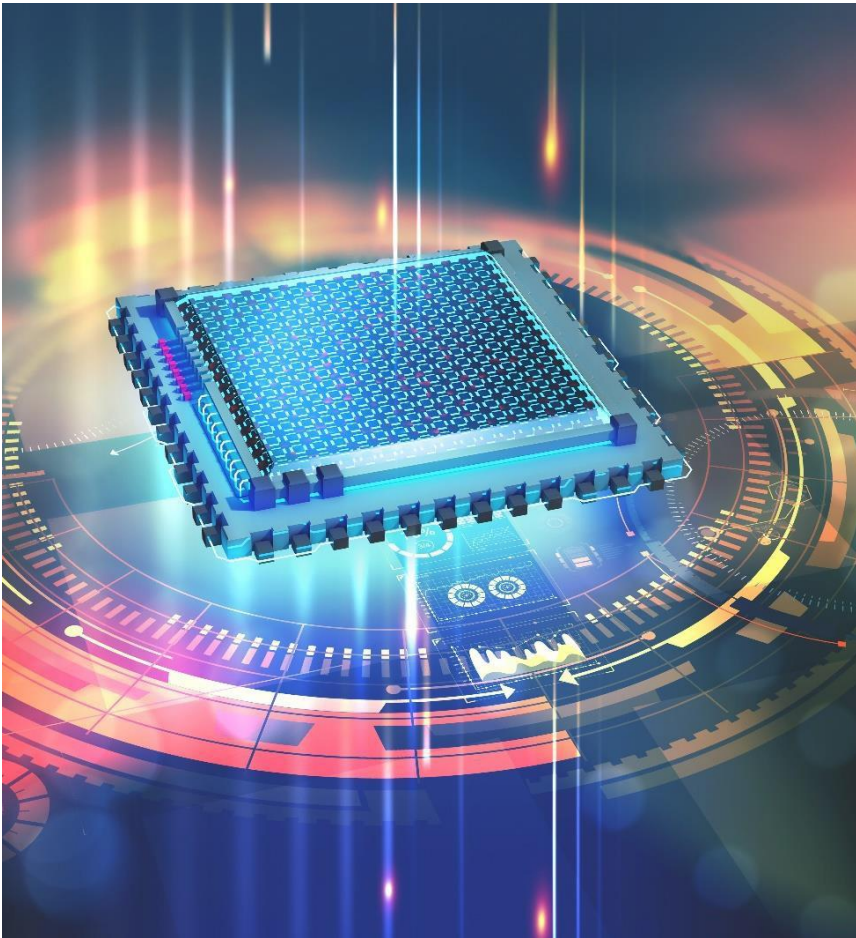


Model RFS-4220T

RF & Microwave Signal Generator



Features

- Wide Frequency Range (100MHz-22GHz)
- Adjustment Level up to +15 dBm
- Low Phase Noise & Excellent Spectral Purity
- Precision Frequency Control
- Desktop & LAN control

Applications

- Automated Test Environment
- Antenna Design
- Aerospace & Defense Research
- Wireless Infrastructure Design
- Satellite Link Testing



Model RFS-4220T
100 MHz – 22 GHz

INTRODUCTION

The RFS-4220T is a compact, high-performance RF and microwave signal generator designed for lab testing, automated systems, and RF development. Offering precise frequency control, low phase noise, and a broad operating range, this generator provides a cost-effective and user-friendly solution for engineers and researchers.

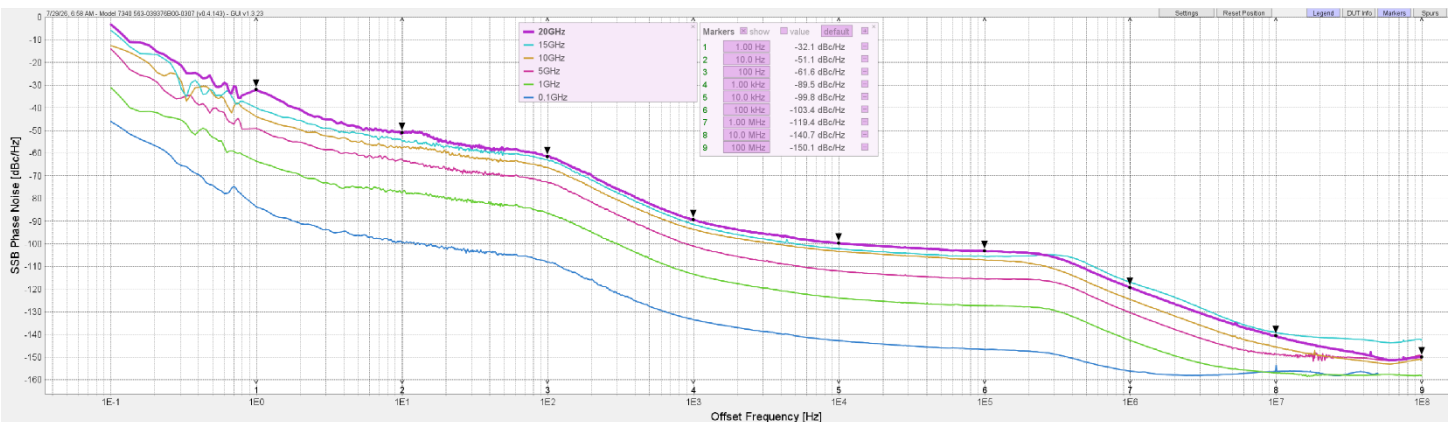
Key Features

- Wide frequency coverage: 0.1 GHz to 22 GHz
- Flexible and user-friendly design: small, portable aluminum enclosure ideal for lab and field use
- High output power: up to +15 dBm with fine power control
- Low phase noise: excellent spectral purity for demanding applications
- Precision frequency control: ultra-fine tuning with small frequency step sizes (<2 Hz or 10 Hz)
- Dual control options: front panel touchscreen and buttons, or remote LAN control

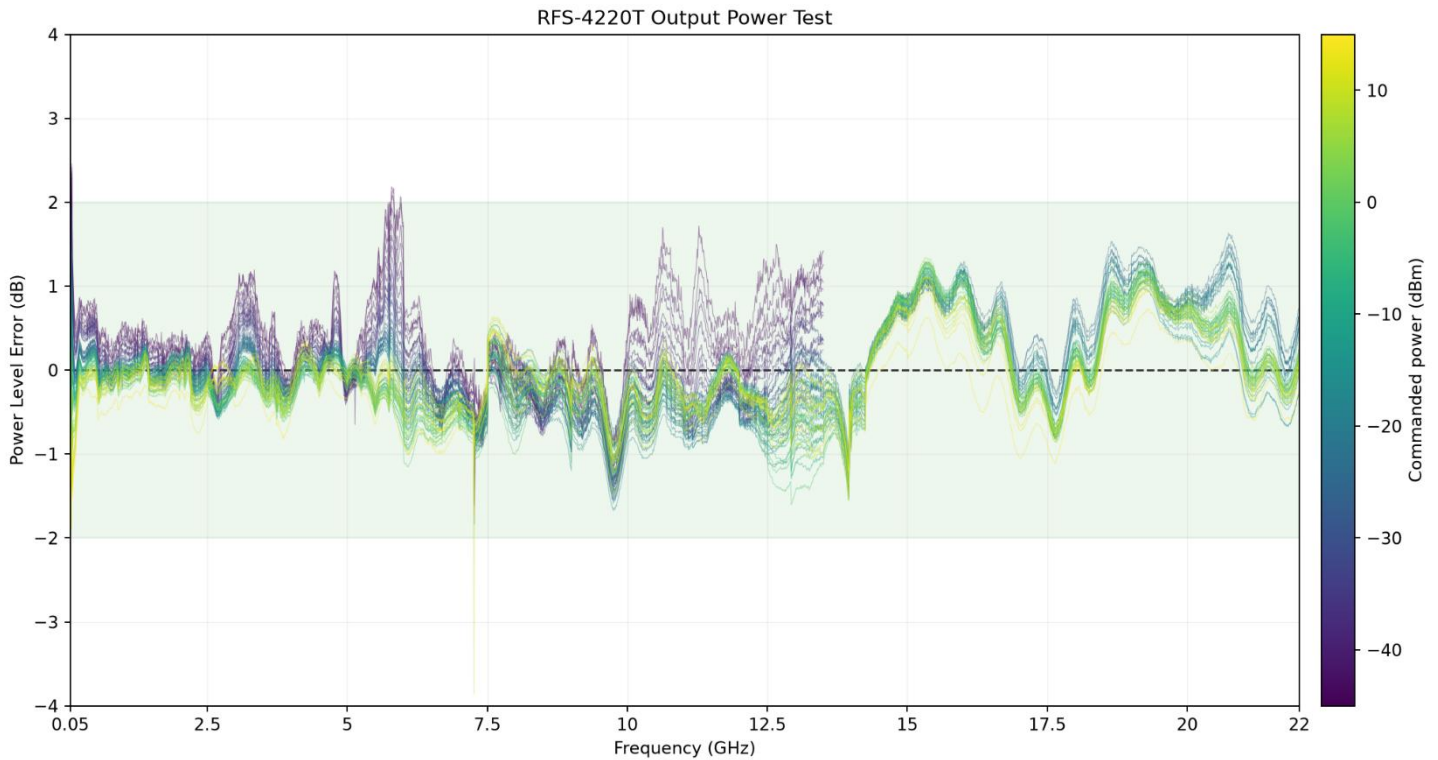
PRODUCT SPECS

Feature	RFS-4220T
Frequency Range	20 MHz – 20 GHz calibrated, settable up to 22 GHz
Power Range	-45 to +15 dBm (0-13.5 GHz) -26 to +15 dBm (13.5-22 GHz)
Power Accuracy	±2.0 dB typical
Frequency Step Size	1 Hz
Harmonic Content	< -25 dBc (typical)
Reference Source	Internal 10.0MHz OCXO or external 10.0 MHz reference
Internal Reference Stability	±10.0 ppb
Control Interface	Touchscreen, front panel buttons, or computer application
Power Supply	90–264 VAC, 50/60 Hz, 100 VA max
Size & Enclosure	8.6 × 4 × 12.25 in (21.84 × 10.16 × 31.12 cm)
Weight	5.6 lbs (2.55 kg)

PHASE NOISE



POWER ACCURACY



APPLICATIONS

- Automated Testing Environments: Suitable for integration into automated test setups
- General RF Lab Use: Ideal for various RF laboratory applications
- Flexible LO Sourcing: Can be used as a local oscillator source in different configurations
- Antenna Design: Useful in the development and testing of antenna systems
- EMC Testing: Applicable in electromagnetic compatibility testing scenarios
- Production Verification and Test Setups: suitable for production line testing and verification processes
- Educational / University Lab Use: beneficial for educational purposes and research in academic settings
- Aerospace / Defense Research: Applicable in aerospace and defense research projects
- 802.11n Development / Testing: Useful in the development and testing of 802.11n wireless standards
- Ku-band Satellite Link Testing: suitable testing Ku-band satellite communication links
- X-Band Radar Applications: applicable in X-band radar system developments
- Up-converting and Down-converting: can be used in frequency conversion applications
- Line of Sight Link Testing: useful in testing line-of-sight communication links
- Wireless Infrastructure Design: beneficial in designing wireless communication infrastructures
- Transponder Verification: suitable for verifying transponder functionalities
- 5G Testing: applicable in the development and testing of 5G technology

Document History

Version	Date	Author	Notes
V1.0	07/21/26	IP	First draft of the datasheet
V1.1	8/3/26	IP	Added phase noise & power accuracy

NOTES

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